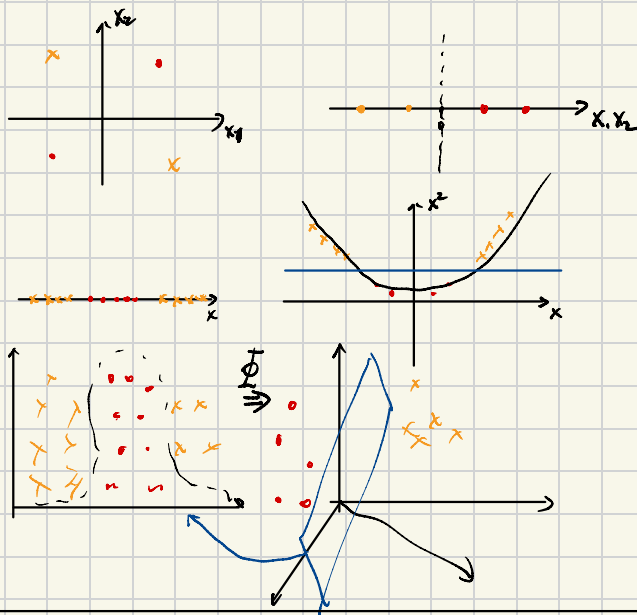


Lecture 10 Non Linear Classifier and Kernels

linear classifier: SVM, LR, Perceptron...

e.g



Idea: transform the input space

- mapping $\Phi: \mathcal{X} \rightarrow \mathcal{Z}$ (high-dim feat. space)
- learn a linear classifier in $\mathcal{Z}$
- $\dim(\mathcal{Z}) > \dim(\mathcal{X})$, $\Rightarrow$ linearly separable
- max the limit $\dim(\mathcal{Z}) \rightarrow \infty$, map each x_i to a func

$$x_i \in \mathbb{R}^d$$

$$\Downarrow$$

$$e^{-\frac{1}{2} \|z - x_i\|^2}$$

kernel SVM

mapping func: $x_i \rightarrow \Phi(x_i)$ (high-dim f.s.)

dual SVM

training: $\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j \Phi(x_i)^T \Phi(x_j)$

s.t. $\sum_i \alpha_i y_i = 0, \alpha_i \geq 0 \forall_i$

Decision Boundary: $w = \sum_i \alpha_i y_i \Phi(x_i)$ $\rightarrow w$ is in $\mathcal{Z}$ space

bias $b = \frac{1}{|\mathcal{S}|} \sum_i y_i - w^T \Phi(x_i) = \frac{1}{|\mathcal{S}|} \sum_i (y_i - \sum_j \alpha_j y_j \Phi(x_j)^T \Phi(x_i))$
 $k(x_j, x_i)$

Decision func: $y_k = \text{sign}(f(\Phi(x_k)))$

given $x_k: f(x_k) = w^T \Phi(x_k) + b = \sum_i \alpha_i y_i \Phi(x_i)^T \Phi(x_k) + b$
 $k(x_i, x_k)$

Note: the entire alg. depends only on $\Phi(x_i)^T \Phi(x_j)$

Define a kernel func.

$$k(x_i, x_j) = \Phi(x_i)^T \Phi(x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$$

- The dual SVM can be re-written w/ kernel func
 $\rightarrow$ non-linear classifier
- we only need the kernel func, not $\Phi(x)$
only define the kernel $\Rightarrow$ kernel trick
- benefits: save time & space explicitly
no need to calculate $\Phi(x)$
- need to compute $O(n^2)$ terms of $k(x_i, x_j)$

$$[k(x_i, x_j)]_{ij} \Rightarrow \text{kernel matrix}$$

Lecture 10 Non Linear Classifier and Kernels

polynomial kernel: $x = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} \in \mathbb{R}^d$, $x' \in \mathbb{R}^d$
(degree-2)

$$k(x, x') = (x^T x')^2 = \left(\sum_{i=1}^d x_i x'_i \right)^2 = \sum_{i=1}^d \sum_{j=1}^d x_i x'_i x_j x'_j$$

$$= \sum_{i,j} x_i x_j \cdot x'_i x'_j$$

$$= \begin{bmatrix} x_1 x_1 & x_1 x_2 & \dots & x_1 x_d & x_2 x_1 & x_2 x_2 & \dots & x_2 x_d & \dots & x_d x_1 & x_d x_2 & \dots & x_d x_d \end{bmatrix} \begin{bmatrix} x'_1 x'_1 \\ x'_1 x'_2 \\ \vdots \\ x'_d x'_1 \\ x'_d x'_2 \\ \vdots \\ x'_d x'_d \end{bmatrix}$$

d^2 entries
 $\Phi(x)^T$

$\Phi(x')$

$$k(x, x') = \underbrace{\Phi(x)^T}_{O(d^2) \text{ multi}} \underbrace{\Phi(x')}_{d^2 \text{ multi}} = (x^T x')^2$$

$\Phi: \mathbb{R}^d \rightarrow \mathbb{R}^{d^2}$

consider decision func

$$f(x) = \sum_i d_i y_i k(x_i, x) + b = \sum_i d_i y_i (x_i^T x)^2 + b$$

$$= \sum_i d_i y_i (x_i^T x) (x_i^T x) + b$$

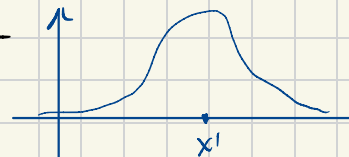
$$= x^T \underbrace{\left[\sum_i d_i y_i x_i x_i^T \right]}_A x + b \rightarrow \text{quadratic func}$$

The kernel func can determine the class of $f(x)$

Gaussian Kernel

$$k(x, x') = e^{-\frac{1}{2\sigma^2} \|x - x'\|^2}$$

Radial basis func. RBF)



- If it is valid (dot-product) kernel?
- If yes, what is $\Phi(x)$?

Lecture 10 Non Linear Classifier and Kernels

Kernel Functions

Kernel trick depends on $K(x, x')$ being a dot-product kernel

Define: a mapping $k: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a dot-product kernel

iff $K(x, x') = \langle \Phi(x), \Phi(x') \rangle$

where $\Phi: \mathcal{X} \rightarrow \mathcal{H}$

$\langle \cdot, \cdot \rangle$ is a dot-product in $\mathcal{H}$

Define: a mapping $K: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a positive definite kernel

iff $\forall n$ and $\forall \{x_1, \dots, x_n\} \subset \mathcal{X}$ (all possible data set samples from $\mathcal{X}$)
the kernel matrix $K = [K(x_i, x_j)]_{i,j}$ is a positive - semi-definite matrix

Define K is a positive - semi-definite matrix iff

- 1) $y^T K y \geq 0 \quad \forall y$, or
- 2) eigenvalues of $K \geq 0$, or
- 3) $K = M M^T$, M has independent columns

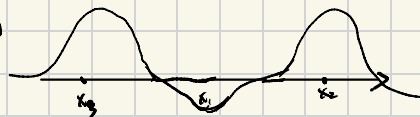
$K(x, x')$ is a dot-product kernel iff it is pos-def kernel

Given a pos-def kernel, what is Φ of $\langle \cdot, \cdot \rangle$?

let $\mathcal{H}$ = space of all linear combination of func $K(\cdot, x_i)$
input x_i fixed

$$\mathcal{H} = \left\{ f(\cdot) \mid f(\cdot) = \sum_{i=1}^m \alpha_i K(\cdot, x_i), \forall m, \forall x_i \in \mathcal{X} \right\}$$

e.g. Gaussian kernel: $K(\cdot, x_i) = e^{-\frac{1}{2\sigma^2} \|\cdot - x_i\|^2}$
func: $f(\cdot)$



$$\text{let: } f(\cdot) = \sum_{i=1}^m \alpha_i K(\cdot, x_i) \quad g(\cdot) = \sum_{j=1}^p \beta_j K(\cdot, x_j)$$

can show dot-product in $\mathcal{H}$

$$\langle f, g \rangle = \sum_i \sum_j \alpha_i \beta_j K(x_i, x_j)$$

e.g. Gaussian kernel: $\langle f, g \rangle = \sum_i \sum_j \alpha_i \beta_j e^{-\frac{1}{2\sigma^2} \|x_i - x_j\|^2}$

special case: $\alpha_i = 1, \beta_j = 1, \alpha_i = 0, \beta_j = 0$

$$\langle f, g \rangle = \langle \underbrace{K(\cdot, x_i)}_{\Phi(x_i)}, \underbrace{K(\cdot, x_j)}_{\Phi(x_j)} \rangle = K(x_i, x_j)$$

mapping

$$x_i \rightarrow \Phi(x_i) = K(\cdot, x_i) \quad \mathbb{R}^d \Rightarrow \mathbb{R}^\infty$$

Lecture 10 Non Linear Classifier and Kernels

$$f(\cdot) = \sum_i \alpha_i k(\cdot, x_i)$$

$$\langle k(\cdot, x), f(\cdot) \rangle = \sum_i \alpha_i k(x_i, x) = f(x)$$

reproducing property: $\sim$ Direct Delta func in conv.

$\mathcal{H}$ is called "reproducing kernel hilbert space" (RKHS)

Representer Theorem

$$\text{Empirical Risk } R_{\text{emp}}(f) = \sum_i L(y_i, f(x_i))$$

Regularizer $\Omega(\|f\|_p)$ $\Omega \geq 0$, strictly monotonically increasing
 $\| \cdot \|_p$ based RKHS & kernel k .

optimization:

$$f^* = \underset{f}{\operatorname{argmin}} R_{\text{emp}}(f) + \lambda \Omega(\|f\|_p)$$

$$\Rightarrow f^* \text{ has form } f^* = \sum_i \alpha_i \underline{k(x, x_i)}, f^* \in \mathcal{H}$$

Note: infinite dim space of f

$\rightarrow$ finite-dim space of α_i , (n -dim)

kernelized ML problem